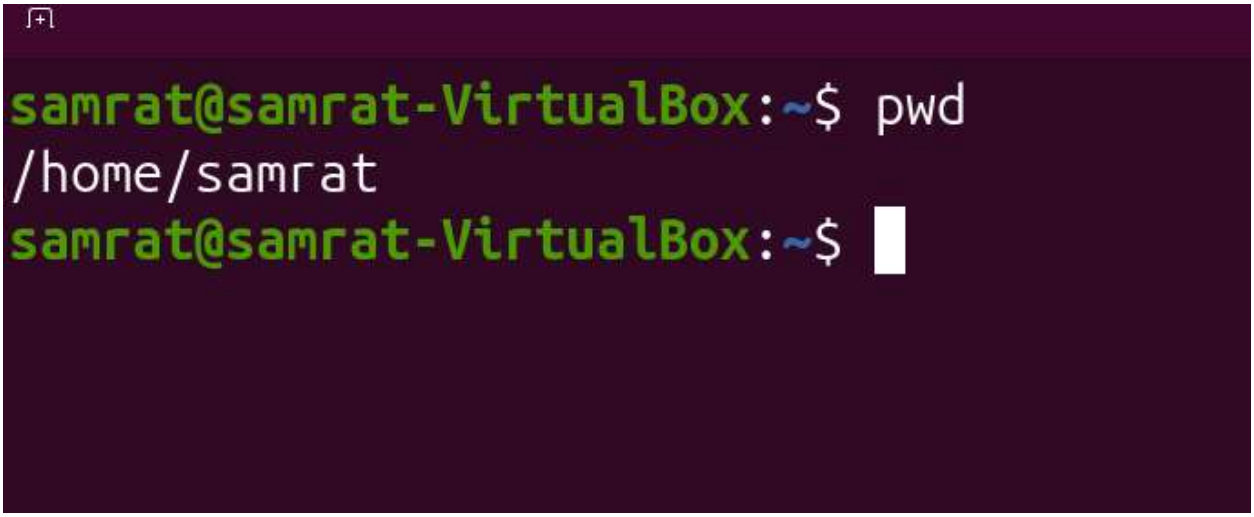
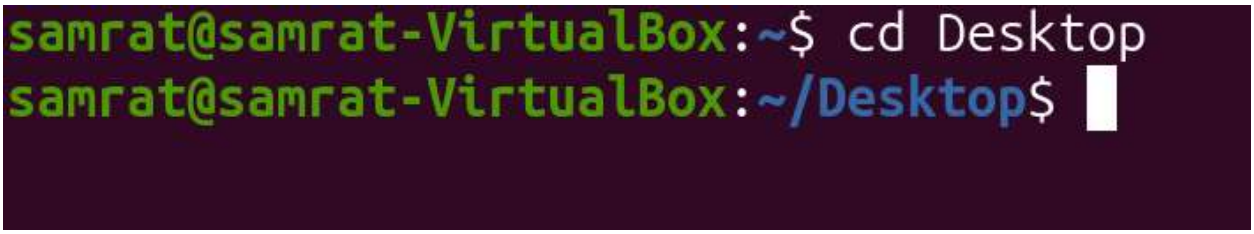

samrat@samrat-VirtualBox:~$ pwd
/home/samrat
samrat@samrat-VirtualBox:~$


samrat@samrat-VirtualBox:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap
samrat@samrat-VirtualBox:~$


samrat@samrat-VirtualBox:~$ cd Desktop
samrat@samrat-VirtualBox:~/Desktop$

```
samrat@samrat-VirtualBox:~/Desktop$ ls -all
total 8
drwxr-xr-x  2 samrat samrat 4096 Aug 12 21:27 .
drwxr-x--- 15 samrat samrat 4096 Jun 16 18:53 ..
samrat@samrat-VirtualBox:~/Desktop$
```

```
samrat@samrat-VirtualBox:~/Desktop$ mkdir oslab
samrat@samrat-VirtualBox:~/Desktop$ tree
.
└── oslab

2 directories, 0 files
samrat@samrat-VirtualBox:~/Desktop$
```

```
samrat@samrat-VirtualBox:~/Desktop$ touch lab33.txt
samrat@samrat-VirtualBox:~/Desktop$ cd ..
samrat@samrat-VirtualBox:~$ tree
.
├── Desktop
│   ├── lab33.txt
│   └── oslab
```

```
samrat@samrat-VirtualBox:~$ vi samrat.txt
samrat@samrat-VirtualBox:~$ echo samrat paudel
samrat paudel
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo apt intsall tree
Error: Invalid operation intsall
samrat@samrat-VirtualBox:~$ sudo apt install tree
tree is already the newest version (2.3.1-1).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                  linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic          linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic   linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 274
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ tree
.
├── Desktop
│   ├── lab33.txt
│   └── oslab
```

```
samrat@samrat-VirtualBox:~$ cp samrat.txt oslab33.txt
samrat@samrat-VirtualBox:~$ tree
.
├── Desktop
│   ├── lab33.txt
│   └── oslab
├── Documents
├── Downloads
├── Music
├── oslab33.txt
├── Pictures
├── Public
└── samrat.txt
```

```
samrat@samrat-VirtualBox:~$ cd ..
samrat@samrat-VirtualBox:/home$
```

```
samrat@samrat-VirtualBox:~$ mkdir -p /home/samrat/Desktop/temp
samrat@samrat-VirtualBox:~$ cp lab33.txt /home/samrat/Desktop/temp/4semlab.txt
samrat@samrat-VirtualBox:~$ tree /home/samrat/Desktop/
/home/samrat/Desktop/
├── lab33.txt
├── oslab
└── temp
    └── 4semlab.txt

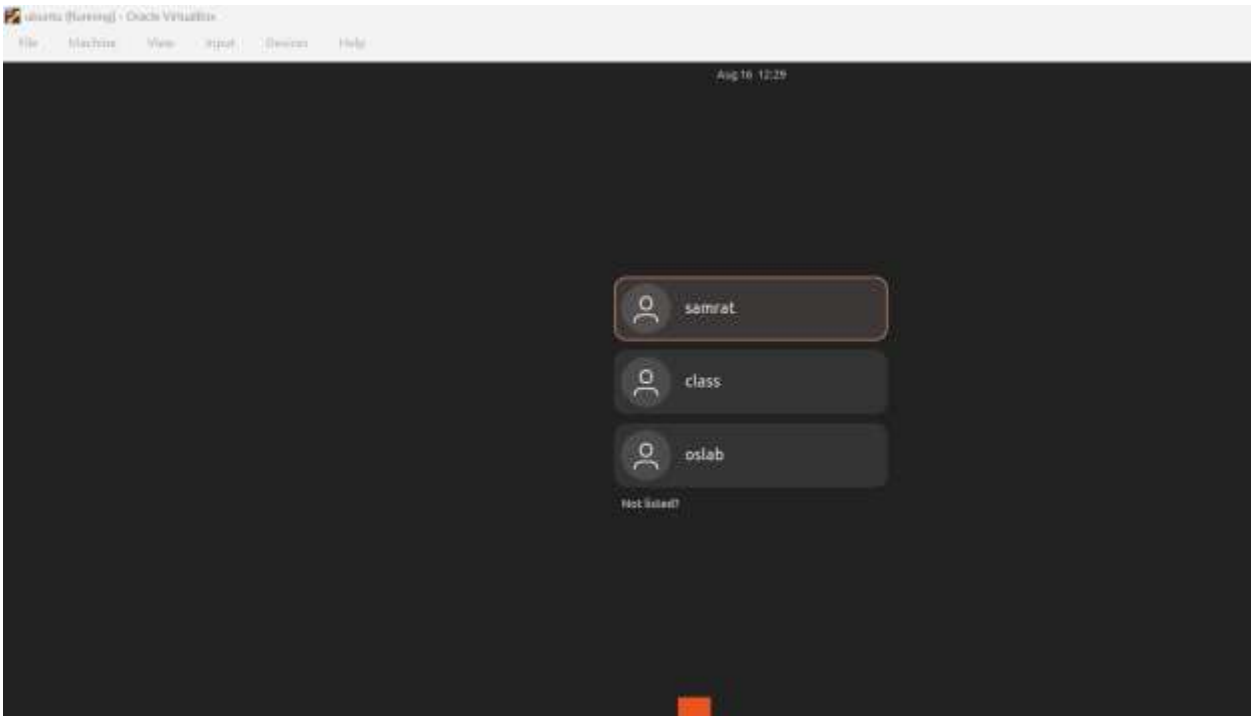
3 directories, 2 files
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ mv oslab33.txt /home/samrat/Desktop/temp/
samrat@samrat-VirtualBox:~$ tree /home/samrat/Desktop/
/home/samrat/Desktop/
├── lab33.txt
├── oslab
└── temp
    ├── 4semlab.txt
    └── oslab33.txt

3 directories, 3 files
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat samrat 4096 Aug 16 12:09 .
```

```
samrat@samrat-VirtualBox:~$ sudo useradd oslab
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
samrat@samrat-VirtualBox:~$ sudo passwd oslab
New password:
Retype new password:
passwd: password updated successfully
samrat@samrat-VirtualBox:~$
```




```
samrat@samrat-VirtualBox:~$ whoami
samrat
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo su -
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
root@samrat-VirtualBox:~# whoami
root
root@samrat-VirtualBox:~#
```

```
samrat@samrat-VirtualBox:~$ sudo groupadd os
samrat@samrat-VirtualBox:~$ getent group os
os:x:1006:
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat samrat 4096 Aug 16 12:09 .
drwxr-xr-x  4 root   root   4096 Jul 10 11:49 ..
-rw-------  1 samrat samrat  940 Aug 12 21:26 .bash_history
-rw-r--r--  1 samrat samrat  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 samrat samrat 3771 Feb 13  2026 .bashrc
drwx------ 14 samrat samrat 4096 Jul 22 10:22 .cache
drwx------ 17 samrat samrat 4096 Jul 10 11:35 .config
drwx------  4 samrat samrat 4096 Jun 16 18:47 .local
-rw-r--r--  1 samrat samrat  807 Feb 13  2026 .profile
drwx------  2 samrat samrat 4096 Jun 16 18:47 .ssh
drwxr-xr-x  4 samrat samrat 4096 Aug 16 12:05 Desktop
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Documents
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Downloads
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Music
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Pictures
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Public
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Templates
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Videos
-rw-rw-r--  1 samrat samrat    0 Aug 16 11:43 samrat.txt
drwx------  7 samrat samrat 4096 Jul  9 21:00 snap
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo usermod -aG lab oslab
samrat@samrat-VirtualBox:~$ groups oslab
oslab : oslab lab
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat samrat 4096 Aug 16 12:09 .
drwxr-xr-x  4 root   root   4096 Jul 10 11:49 ..
-rw-------  1 samrat samrat  940 Aug 12 21:26 .bash_history
-rw-r--r--  1 samrat samrat  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 samrat samrat 3771 Feb 13  2026 .bashrc
drwx------ 14 samrat samrat 4096 Jul 22 10:22 .cache
drwx------ 17 samrat samrat 4096 Jul 10 11:35 .config
drwx------  4 samrat samrat 4096 Jun 16 18:47 .local
-rw-r--r--  1 samrat samrat  807 Feb 13  2026 .profile
drwx------  2 samrat samrat 4096 Jun 16 18:47 .ssh
drwxr-xr-x  4 samrat samrat 4096 Aug 16 12:05 Desktop
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Documents
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Downloads
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Music
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Pictures
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Public
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Templates
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Videos
-rw-rw-r--  1 samrat samrat    0 Aug 16 11:43 samrat.txt
drwx------  7 samrat samrat 4096 Jul  9 21:00 snap
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo chown -R samrat:oslab .
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat oslab 4096 Aug 16 12:09 .
drwxr-xr-x  4 root   root  4096 Jul 10 11:49 ..
-rw-------  1 samrat oslab  940 Aug 12 21:26 .bash_history
-rw-r--r--  1 samrat oslab  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 samrat oslab 3771 Feb 13  2026 .bashrc
drwx------ 14 samrat oslab 4096 Jul 22 10:22 .cache
drwx------ 17 samrat oslab 4096 Jul 10 11:35 .config
drwx------  4 samrat oslab 4096 Jun 16 18:47 .local
-rw-r--r--  1 samrat oslab  807 Feb 13  2026 .profile
drwx------  2 samrat oslab 4096 Jun 16 18:47 .ssh
drwxr-xr-x  4 samrat oslab 4096 Aug 16 12:05 Desktop
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Documents
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Downloads
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Music
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Pictures
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Public
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Templates
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Videos
-rw-rw-r--  1 samrat oslab    0 Aug 16 11:43 samrat.txt
drwx------  7 samrat oslab 4096 Jul  9 21:00 snap
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ zip archive.zip oslab33.txt
  adding: oslab33.txt (stored 0%)
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ unzip archive.zip
Archive:  archive.zip
replace oslab33.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: N
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ ln -s /home/samrat/Desktop/oslab/lab33.txt /home/samrat/Desktop/
ln: Already exists
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ cat lab33.txt
samrat
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
tmpfs             695724     2064    693660   1% /run
/dev/sda2       55677136 11275616  41540876  22% /
tmpfs            1739308        0   1739308   0% /dev/shm
none                1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            1739312        8   1739304   1% /tmp
none                1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             347860       84    347776   1% /run/user/1000
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ du -sh *
4.0K    archive.zip
4.0K    lab33.txt
0       oslab33.txt
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ hostname
samrat-VirtualBox
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ time sleep 3

real    0m3.138s
user    0m0.015s
sys     0m0.025s
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ ping google.com
PING google.com (142.251.126.113) 56(84) bytes of data.
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=1 ttl=64 time=61.0 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=2 ttl=64 time=54.6 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=3 ttl=64 time=50.6 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=4 ttl=64 time=68.4 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=5 ttl=64 time=56.4 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=6 ttl=64 time=59.0 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=7 ttl=64 time=71.2 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=8 ttl=64 time=49.5 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=9 ttl=64 time=55.8 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=10 ttl=64 time=50.4 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=11 ttl=64 time=51.2 ms
```

```
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI> python greedybestsearch.py
goal node could be reached
goal node could be reached
goal node could be reached
goal node could be reached
goal node could be reached
found goal
Initial State: A
Goal State: F
Path Found by Greedy Best-First Search: ['A', 'C', 'F']
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI>
```

```
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI> python lab5.py
Graph adjacency list with costs:
A -> [('B', 1), ('C', 4), ('D', 3)]
B -> [('E', 2)]
C -> [('E', 1), ('F', 5)]
D -> [('F', 6)]
E -> [('F', 1)]
F -> []

Heuristic values:
h(A) = 7
h(B) = 6
h(C) = 2
h(D) = 4
h(E) = 1
h(F) = 0

Start Node: A
Goal Node: F
Path found: A -> C -> E -> F
Total cost: 6
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI>
```

```
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI> python minmax.py
Leaf values: [3, 5, 4, 5]
Max depth: 2

The optimal value is: 4
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI>
```

```
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI> python waterjug.py
Jug A Capacity: 5 liters
Jug B Capacity: 3 liters
Initial State: (0, 0)
Goal State: Jug A contains 1 liter -> (1, _)

Solution found! Step-by-step jug contents:

Step 0: Jug A = 0, Jug B = 0
Step 1: Jug A = 0, Jug B = 3
Step 2: Jug A = 3, Jug B = 0
Step 3: Jug A = 3, Jug B = 3
Step 4: Jug A = 5, Jug B = 1
Step 5: Jug A = 0, Jug B = 1
Step 6: Jug A = 1, Jug B = 0

Goal reached in 6 steps.
PS C:\Users\ACER\OneDrive\Desktop\4th sem\AI>
```